

iForest Algorithm

Initial Dataset

1. $[-41, 28, 16]$
2. $[-6, -7, 36]$
3. $[-42, 20, -30]$
4. $[-41, 3, 48]$
5. $[24, 26, 22]$
6. $[29, 1, -38]$
7. $[34, -5, 0]$
8. $[-13, -32, ~~4~~ 43]$
9. $[28, 15, -10]$
10. $[33, 5, -6]$
11. $[-5, -28, -41]$
12. $[6, 39, -44]$
13. $[36, 33, -23]$
14. $[13, -34, 26]$
15. $[20, -15, -44]$.

Subsample index for 5 trees

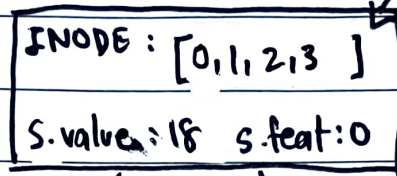
1. $[6, 1, 9, 10]$
2. $[8, 2, 1, 7]$
3. $[8, 7, 10, 1]$
4. $[4, 11, 2, 12]$
5. $[3, 13, 5, 6]$.

Creating Trees

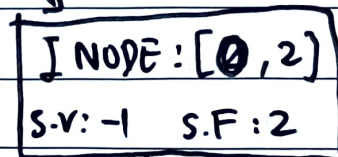
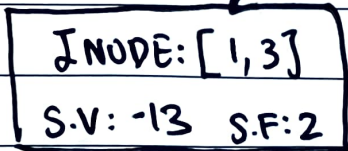
Tree 1:

Depth
0

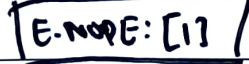
Data pt index for the subsample
taken from sample index or
sample



1

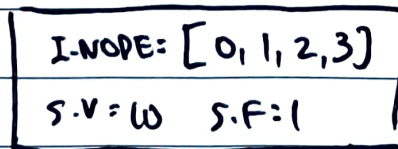


2

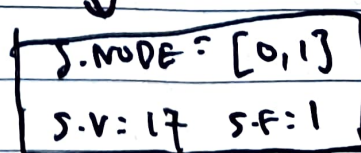
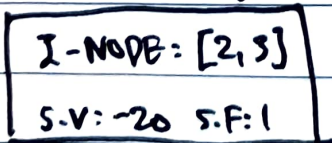


Depth
0

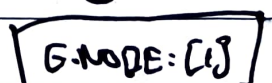
Tree 2:



1

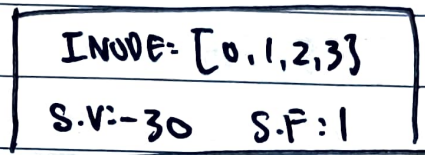


2

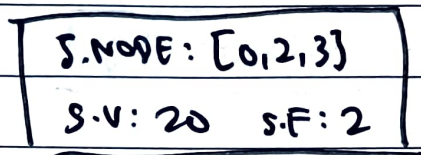


Depth
0

Tree 3:



1



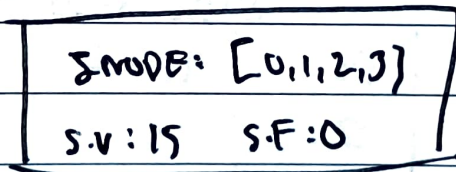
2



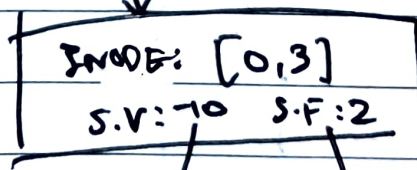
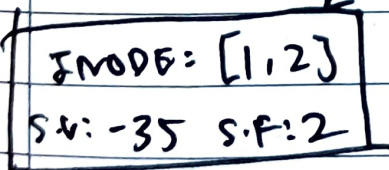
depth

Tree 4:

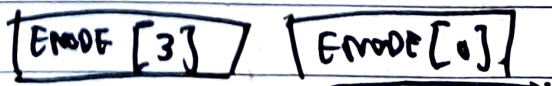
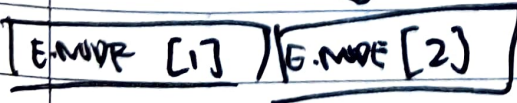
0



1

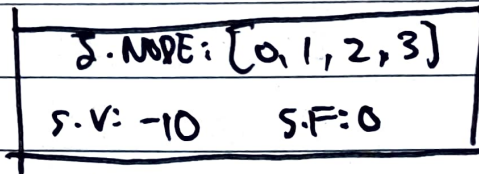


2

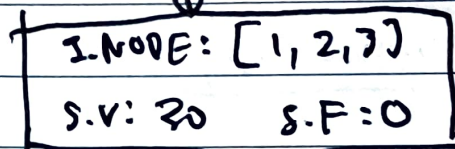
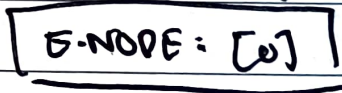


depth
0

Tree 5



1



2.



Path length

	Test set	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5
1.	[-41, 28, 16].	2	2	3	2	1
2.	[-6, -7, 36]	2	2	2	2	3
3.	[-42, 20, -30]	2	2	3	2	1
4.	[-41, 3, 48]	2	2	2	2	1
5.	[24, 26, 22].	2	2	2	2	3
6.	[29, 1, -38].	2	2	3	2	3
7.	[24, -5, 0].	2	2	3	2	2
8.	[-13, -32, 43].	2	2	1	2	1
9.	[28, 15, -10].	2	2	3	2	3
10.	[33, 5, -6]	2	2	3	2	2

$$2^{-\frac{E[h(x)]}{c(n)}}$$

2

Avg path length

$S(x, n)$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

$$c(4) = 2H(3) - \frac{2(3)}{4}$$

$$\approx \underline{\underline{2.1667}}$$

1.	2.0	0.527
2.	2.2	0.495
3.	2.0	0.527
4.	1.8	0.562
5.	2.2	0.495
6.	2.4	0.464
7.	2.2	0.495
8.	1.6	0.599
9.	2.4	0.464
10.	2.2	0.495

LARGEST.